

Digital Twins for Predictive Maintenance in Automotive Aftersales

Oliver Feng
Wirtschaftsinformatik
DHBW Stuttgart
Stuttgart, Germany
wi23030@lehre.dhbw-stuttgart.de

Maksim Bogachenkov
Wirtschaftsinformatik
DHBW Stuttgart
Stuttgart, Germany
wi23024@lehre.dhbw-stuttgart.de

Joel Yeraí Martínez Campo
Wirtschaftsinformatik
DHBW Stuttgart
Stuttgart, Germany
wi23194@lehre.dhbw-stuttgart.de

Abstract—TODO: Abstract here.

Index Terms—Digital Twin, Predictive Maintenance, Industry 4.0, Machine Learning, IoT, Condition Monitoring

I. INTRODUCTION

- Hook
- Smart to Wise Bezug
- Problemstellung
- Zeilsetzung
- RQ, Forschungslücke
- Beitrag der Arbeit
- Aufbau der Arbeit

II. RELATED WORKS & THEORETICAL BACKGROUND

A. Digital Twins

B. Vehicle Maintenance Strategies

Vehicle maintenance strategies form a maturity hierarchy: reactive maintenance repairs only after failure [1]; preventive maintenance schedules service at fixed mileage or time intervals, leading to either premature replacement or undetected wear [1], [2]; condition-based maintenance (CBM) couples the service decision to sensor thresholds [2]; and predictive maintenance (PdM) extends CBM by forecasting the future evolution of degradation, estimating the remaining time or distance until end-of-life is reached [1], [3].

PdM relies on three data layers: on-board sensors delivering high-frequency signals such as vibration, current, voltage, and cell temperature [2]–[4]; the On-Board Diagnostic (OBD) interface providing standardized diagnostic codes [5], [6]; and over-the-air (OTA) telematics enabling continuous low-latency exchange with the backend [4], [6], [7]. Together, these layers transform the vehicle from a periodically inspected asset into a continuously observed system, a shift first documented by Zhang et al. [5] in their work on Connected Vehicle Diagnostics and Prognostics at General Motors.

PdM at this scale is not a stand-alone technique but a service layer built on a digital twin (DT) of the vehicle. As Lopes et al. [7] and Hadraoui et al. [2] describe, the DT supplies the synchronized virtual representation against which sensor streams are interpreted, anomalies detected, and remaining-useful-life estimates computed. Without a DT, telemetry must be processed in isolation; without PdM, the DT remains a

passive monitor. In current automotive DT architectures, predictive analytics is therefore typically positioned as a dedicated microservice within the DT's analytics core [1], [4], [7].

C. Machine Learning in PdM

Three problem classes structure ML-based PdM. Classification assigns sensor patterns to discrete fault categories using Support Vector Machines, Random Forests, or Convolutional Neural Networks on time-frequency representations of vibration signals [1], [3]. Regression targets Remaining Useful Life (RUL) estimation, dominated by Long Short-Term Memory (LSTM) networks for their ability to model long-range temporal dependencies in degradation signals [1], [4]. Anomaly detection operates without explicit labels, learning nominal behavior and flagging deviations as failure precursors [1], [4]. AbouGrad et al. report classification accuracies of approximately 89% for vibration-based fault detection and 92% for LSTM-based RUL estimation in automotive contexts, though they emphasize that these figures depend strongly on dataset quality and operating conditions.

A recurring constraint is data: AbouGrad et al. and Hadraoui et al. note that preprocessing consumes 70–90% of project time, genuine failure data is structurally scarce, and public benchmarks representative of in-service behavior are largely absent. Model-based and hybrid approaches that embed physical degradation laws into the learning pipeline are therefore preferred over purely data-driven ones, as they extrapolate more reliably under data scarcity [2], [3], [8]. Beyond accuracy, deployment in safety-relevant contexts raises the question of explainability: deep architectures achieve high accuracy but operate as opaque models whose decisions cannot readily be justified to engineers, customers, or regulators [1]. Explainable AI (XAI) techniques such as feature attribution and attention-based saliency are increasingly integrated into PdM pipelines to expose the reasoning behind a prediction, marking the conceptual bridge from smart to wise information systems [1].

D. Automotive Aftersales & Customer Lifecycle

After-sales encompasses all post-purchase activities—installation, repair, maintenance, quality control, and complaint handling—that help customers access supplier services and resolve product issues [9]. In the automotive context, it

functions as a dedicated business module bundling maintenance, repair, and warranty offerings to maximise product use throughout the asset lifecycle while meeting financial and customer-performance targets [10]. Building post-sales relationships strengthens customer loyalty, yields actionable insights, and creates upselling and cross-selling opportunities [11], making the automotive aftersales market increasingly attractive as demand for higher-quality service expands [10].

Historically, aftersales was predominantly *reactive*: service was initiated only after a customer reported a failure, constraining both the timeliness and the breadth of interventions. Verstrepen et al. [9] observe that maintenance-free vehicles tend to disengage customers after warranty expiry and conclude that improved information systems and service processes are essential to retain them. The literature records a clear evolutionary shift in the automotive industry—from reactive, failure-based maintenance toward proactive and predictive strategies—driven by growing recognition that anticipating issues before they arise improves performance, efficiency, and long-term operational excellence [12].

The contemporary aftersales paradigm is therefore increasingly *proactive*. Challagalla et al. [13] demonstrate that Proactive Post-Sales (PPS) service improves customer satisfaction and retention by enabling suppliers to identify and resolve issues before customers encounter them, yielding more productive interactions, broader customer reach, and greater innovation potential compared with the reactive approach. Proactive services must be tailored to customer needs and channel workload, necessitating tight integration of operational and informational data [9]. González-Prida et al. [14] provide a multiphase framework for deploying IoT, digital twins, and AI to this end, showing that machine learning and digital-twin-based approaches enable predictive interventions and better asset lifecycle optimisation, with concrete benefits including cost reduction, improved customer satisfaction, and extended asset lifetimes. Digital twins in particular allow maintenance to be grounded in current asset data, supporting precise, personalised interventions that extend operational lifetimes and improve service transparency [14]. To ensure that AI-driven service decisions remain trustworthy, the OECD AI Principles require that aftersales systems be transparent, safe, explainable, and non-discriminatory, so that automated failure predictions and warranty decisions remain fair, auditable, and privacy-aware [15].

Forschungslücke

While these contributions establish the technical foundations of DT-based PdM, existing work concentrates either on algorithmic aspects in isolation [1], [3] or on industrial and aerospace applications [8]. The integration of DT-based PdM into automotive aftersales business models — including its implications for customer interaction, upselling potential, and vehicle lifecycle value — remains underexplored. The present work addresses this gap.

III. METHODOLOGY

- LR

- Illustrative Case Study
- (DSR)

IV. REFERENCE ARCHITECTURE

- Sensorik
- ML-Algorithmen
- Kennzahlen
- Schnittstellen
- Datenmanagement

V. ILLUSTRATIVE CASE STUDY

- A. *Lifecycle Comparison*
- B. *Assumption Framework*
- C. *Illustrative Calculation & Results*
- D. *Limitations*

VI. DISCUSSION AND RECOMMENDATIONS

- Interpretation der Ergebnisse
- Implikationen fuer Forschung und Praxis

VII. CONCLUSION & FUTURE WORK

- Zusammenfassung der wichtigsten Erkenntnisse
- Beitrag der Arbeit
- Ausblick auf weitere Forschung [?]

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ERKLÄRUNG ZUR NUTZUNG VON KI-TOOLS

TODO: KI-Erklärung hier einfügen (gemaess DHBW-Vorgabe).